

Conditional Statements: Making Trading Decisions

Imagine you're a trader watching the market. You have a simple strategy: if a stock's price drops below a certain threshold, you buy it, believing it's undervalued. If it rises above another threshold, you sell, taking your profits.

This is exactly the kind of decision-making that conditional statements in Python enable. They allow your program to check conditions and act accordingly—just like a trader would, but without emotional biases or fatigue.

The if Statement: Your First Decision Point

The if statement is the most basic form of conditional logic. It simply says: "If this condition is true, then do something."

Here's a simple example:

```
1. price = 150
2. buy_threshold = 145
3.
4. if price < buy_threshold:
5.     print("Buy signal generated!")
```

In this code, Python checks if the current price (150) is less than our buy threshold (145). Since 150 is not less than 145, the condition is false, and Python skips the indented code block.

Let's see a more realistic trading example:

```
1. stock_price = 273.20
2. moving_average_50 = 265.40
3. volume = 3200000
4. average_volume = 2800000
5.
6. if (stock_price > moving_average_50) and (volume > average_volume):
7.     print("Bullish signal: Price above 50-day MA with above-average volume")
```